

Labwork 4

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1

Welcome x postgres/postgres@PostgreSQL 18* x

postgres/postgres@PostgreSQL 18

Query Query History AI Assistant Scratch Pad x

```
1 -- 1. Airline names in uppercase
2 SELECT UPPER(airline_name) AS airline_name_upper
3 FROM airline;
4
5 -- 2. Replace "Air" with "Aero" in airline names
6 SELECT airline_name,
7        REPLACE(airline_name, 'Air', 'Aero') AS new_name
8 FROM airline;
9
10 -- 3. Flight numbers common to airline 1 and airline 2
11 SELECT flight_id AS flight_number
12 FROM flights
13 WHERE airline_id = 1
14 INTERSECT
15 SELECT flight_id
16 FROM flights
17 WHERE airline_id = 2;
18
19 -- 4. Airports with "Regional" and "Air" in the name
20 SELECT *
21 FROM airport
22 WHERE airport_name ILIKE '%Regional%'
```

Data Output Messages Notifications

Showing rows: 1 to 204 Page No: 1 of 1

	airline_name_upper text
1	AIRLINE 107
2	AIRLINE 108
3	AIRLINE 109
4	AIRLINE 110
5	AIRLINE 111
6	AIRLINE 112
7	AIRLINE 113
8	AIRLINE 114
9	AIRLINE 115
10	AIRLINE 116
11	AIRLINE 117
12	AIRLINE 118
13	AIRLINE 119
14	AIRLINE 120
15	AIRLINE 121

Total rows: 204 Query complete 00:00:01.108 LF Ln 1, Col 1

2

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Query Query History AI Assistant

Execute script F5

```
-- 1. Airline names in uppercase
SELECT UPPER(airline_name) AS airline_name_upper
FROM airline;

-- 2. Replace "Air" with "Aero" in airline names
SELECT airline_name,
       REPLACE(airline_name, 'Air', 'Aero') AS new_name
FROM airline;

-- 3. Flight numbers common to airline 1 and airline 2
SELECT flight_id AS flight_number
FROM flights
WHERE airline_id = 1
INTERSECT
SELECT flight_id
FROM flights
WHERE airline_id = 2;

-- 4. Airports with "Regional" and "Air" in the name
SELECT *
FROM airport
WHERE airport_name ILIKE '%Regional%'
```

Data Output Messages Notifications

Showing rows: 1 to 204 Page No: 1 of 1

	airline_name character varying (50)	new_name text
1	Airline 107	Aeroline 107
2	Airline 108	Aeroline 108
3	Airline 109	Aeroline 109
4	Airline 110	Aeroline 110
5	Airline 111	Aeroline 111
6	Airline 112	Aeroline 112
7	Airline 113	Aeroline 113
8	Airline 114	Aeroline 114
9	Airline 115	Aeroline 115
10	Airline 116	Aeroline 116
11	Airline 117	Aeroline 117
12	Airline 118	Aeroline 118
13	Airline 119	Aeroline 119
14	Airline 120	Aeroline 120
15	Airline 121	Aeroline 121

Total rows: 204 Query complete 00:00:01.122

Successfully run. Total query runtime: 1 secs 122 msec. 204 rows affected.

LF Ln 5, Col 1

3

The screenshot shows a PostgreSQL IDE interface with a dark theme. The main editor displays four SQL queries:

```
1 -- 1. Airline names in uppercase
2 SELECT UPPER(airline_name) AS airline_name_upper
3 FROM airline;
4
5 -- 2. Replace "Air" with "Aero" in airline names
6 SELECT airline_name,
7        REPLACE(airline_name, 'Air', 'Aero') AS new_name
8 FROM airline;
9
10 -- 3. Flight numbers common to airline 1 and airline 2
11 SELECT flight_id AS flight_number
12 FROM flights
13 WHERE airline_id = 1
14 INTERSECT
15 SELECT flight_id
16 FROM flights
17 WHERE airline_id = 2;
18
19 -- 4. Airports with "Regional" and "Air" in the name
20 SELECT *
21 FROM airport
22 WHERE airport_name ILIKE '%Regional%'
```

A tooltip labeled "Execute script" with the "F5" key icon is positioned over the first query. The interface includes a "Scratch Pad" on the right and a "Data Output" section at the bottom. The "Data Output" section shows a table with one column, "flight_number", and one row with the value "integer". A green status bar at the bottom right indicates: "✓ Successfully run. Total query runtime: 999 msec. 0 rows affected. ✕". The bottom status bar shows "Total rows: 0", "Query complete 00:00:01.166", and "Ln 10, Col 1".

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Query

Query History

AI Assistant

Scratch Pad x

5

-- 2. Replace "Air" with "Aero" in airline names

6

SELECT airline_name,

7

REPLACE(airline_name, 'Air', 'Aero') AS new_name

8

FROM airline;

9

10

-- 3. Flight numbers common to airline 1 and airline 2

11

SELECT flight_id AS flight_number

12

FROM flights

13

WHERE airline_id = 1

14

INTERSECT

15

SELECT flight_id

16

FROM flights

17

WHERE airline_id = 2;

18

19

-- 4. Airports with "Regional" and "Air" in the name

20

SELECT *

21

FROM airport

22

WHERE airport_name ILIKE '%Regional%'

23

AND airport_name ILIKE '%Air%';

24

25

-- 5. Passenger names with birth date as 'Month DD, YYYY'

26

SELECT first_name || ' ' || last_name AS passenger_name,

27

TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS birth_date_formatted;

Data Output

Messages

Notifications

SQL

airport_id [PK] integer

airport_name character varying (50)

country character varying (50)

state character varying (50)

city character varying (50)

created_at timestamp without time zone

updated_at timestamp without time zone

Successfully run. Total query runtime: 1 secs 14 msec. 0 rows affected.

Total rows: 0

Query complete 00:00:01.014

Ln 19, Col 1

5

The screenshot shows a PostgreSQL IDE interface with a dark theme. The main window displays a SQL query with line numbers 15 through 36. The query includes several SQL statements: an `INTERSECT` clause, a `SELECT` from `flights` with a `WHERE` clause, a `SELECT *` from `airport` with `WHERE` and `AND` conditions, a `SELECT` statement for passenger names and birth dates, and two more `SELECT` statements for flight numbers and delayed flights. The query is highlighted in a light blue box. Below the query editor, the 'Data Output' tab is active, showing a table with two columns: `passenger_name` and `birth_date_formatted`, both of type `text`. The table is currently empty. At the bottom of the IDE, a status bar indicates 'Total rows: 0' and 'Query complete 00:00:00.928'. A green notification bar at the bottom right states 'Successfully run. Total query runtime: 928 msec. 0 rows affected.'

```
15 INTERSECT
16 SELECT flight_id
17 FROM flights
18 WHERE airline_id = 2;
19
20 -- 4. Airports with "Regional" and "Air" in the name
21 SELECT *
22 FROM airport
23 WHERE airport_name ILIKE '%Regional%'
24 AND airport_name ILIKE '%Air%';
25
26 -- 5. Passenger names with birth date as 'Month DD, YYYY'
27 SELECT first_name || ' ' || last_name AS passenger_name,
28        TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS birth_date_formatted
29 FROM passengers;
30
31 -- 6. Flight numbers delayed (by actual arrival time)
32 SELECT DISTINCT flight_id AS flight_number
33 FROM flights
34 WHERE act_arrival_time IS NOT NULL
35 AND act_arrival_time > sch_arrival_time;
36
37 -- 7. Flights that arrived late, with the delay
```

passenger_name	birth_date_formatted
----------------	----------------------

Successfully run. Total query runtime: 928 msec. 0 rows affected.

Total rows: 0 | Query complete 00:00:00.928 | LF | Ln 25, Col 1

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Query Query History AI Assistant

Scratch Pad x

19 -- 4. Airports with "Regional" and "Air" in the name

20 SELECT *

21 FROM airport

22 WHERE airport_name ILIKE '%Regional%'

23 AND airport_name ILIKE '%Air%';

24

25 -- 5. Passenger names with birth date as 'Month DD, YYYY'

26 SELECT first_name || ' ' || last_name AS passenger_name,

27 TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS birth_date_formatted

28 FROM passengers;

29

30 -- 6. Flight numbers delayed (by actual arrival time)

31 SELECT DISTINCT flight_id AS flight_number

32 FROM flights

33 WHERE act_arrival_time IS NOT NULL

34 AND act_arrival_time > sch_arrival_time;

35

36 -- 7. Flights that arrived late, with the delay

37 SELECT flight_id, sch_arrival_time, act_arrival_time,

38 act_arrival_time - sch_arrival_time AS delay

39 FROM flights

40 WHERE act_arrival_time > sch_arrival_time

41 ORDER BY delay DESC;

Data Output Messages Notifications

flight_number integer

Total rows: 0 Query complete 00:00:00.936

Successfully run. Total query runtime: 936 msec. 0 rows affected.

LF Ln 30, Col 1

7

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postgres/postgres@PostgreSQL 18

Query Query History AI Assistant Scratch Pad X

```
27 TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS birth_date_formatted
28 FROM passengers;
29
30 -- 6. Flight numbers delayed (by actual arrival time)
31 SELECT DISTINCT flight_id AS flight_number
32 FROM flights
33 WHERE act_arrival_time IS NOT NULL
34 AND act_arrival_time > sch_arrival_time;
35
36 -- 7. Flights that arrived late, with the delay
37 SELECT flight_id, sch_arrival_time, act_arrival_time,
38        act_arrival_time - sch_arrival_time AS delay
39 FROM flights
40 WHERE act_arrival_time > sch_arrival_time
41 ORDER BY delay DESC;
42
43 -- 8. Airlines from France/Portugal/Poland created in the period
44 SELECT *
45 FROM airline
46 WHERE airline_country IN ('France', 'Portugal', 'Poland')
47 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
48
49 -- 9. Top 3 overweight baggage (> 25 kg)
```

Data Output Messages Notifications

flight_id	sch_arrival_time	act_arrival_time	delay
[PK] integer	timestamp without time zone	timestamp without time zone	interval

Successfully run. Total query runtime: 1 secs 160 msec. 0 rows affected.

Total rows: 0 Query complete 00:00:01.160 LF Ln 36, Col 1

8

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postgres/postgres@PostgreSQL 18

Query Query History AI Assistant

Scratch Pad X

```
37 SELECT flight_id, sch_arrival_time, act_arrival_time,
38       act_arrival_time - sch_arrival_time AS delay
39 FROM flights
40 WHERE act_arrival_time > sch_arrival_time
41 ORDER BY delay DESC;
42
43 -- 8. Airlines from France/Portugal/Poland created in the period
44 SELECT *
45 FROM airline
46 WHERE airline_country IN ('France', 'Portugal', 'Poland')
47       AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
48
49 -- 9. Top 3 overweight baggage (> 25 kg)
50 SELECT *
51 FROM baggage
52 WHERE weight_in_kg > 25
53 ORDER BY weight_in_kg DESC
54 LIMIT 3;
55
56 -- 10. Youngest passenger(s), full name
57 SELECT first_name || ' ' || last_name AS full_name, date_of_birth
58 FROM passengers
59 WHERE date_of_birth = (SELECT MAX(date_of_birth) FROM passengers);
```

Data Output Messages Notifications

airline_id [PK] integer

airline_code character varying (30)

airline_name character varying (50)

airline_country character varying (50)

created_at timestamp without time zone

updated_at timestamp without time zone

Successfully run. Total query runtime: 1 secs 90 msec. 0 rows affected.

Total rows: 0 Query complete 00:00:01.090

LF Ln 43, Col 1

9

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Query Query History AI Assistant

Scratch Pad x

```
-- 8. Airlines from France/Portugal/Poland created in the period
SELECT *
FROM airline
WHERE airline_country IN ('France', 'Portugal', 'Poland')
AND created_at BETWEEN '2023-11-01' AND '2024-03-31';

-- 9. Top 3 overweight baggage (> 25 kg)
SELECT *
FROM baggage
WHERE weight_in_kg > 25
ORDER BY weight_in_kg DESC
LIMIT 3;

-- 10. Youngest passenger(s), full name
SELECT first_name || ' ' || last_name AS full_name, date_of_birth
FROM passengers
WHERE date_of_birth = (SELECT MAX(date_of_birth) FROM passengers);

-- 11. Cheapest booking price on each booking platform
SELECT booking_platform, MIN(ticket_price) AS cheapest_price
FROM booking
GROUP BY booking_platform;
```

Data Output Messages Notifications

baggage_id [PK] integer

weight_in_kg numeric (4,2)

created_at timestamp without time zone

updated_at timestamp without time zone

booking_id integer

Total rows: 0 Query complete 00:00:01.178

Successfully run. Total query runtime: 1 secs 178 msec. 0 rows affected.

LF Ln 49, Col 1

10

The screenshot shows a PostgreSQL IDE interface with a dark theme. The main window displays a SQL query with line numbers 47 through 69. The query includes comments for queries 9, 10, 11, and 12. Query 10 is highlighted in blue. The query is as follows:

```
47      AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
48
49  -- 9. Top 3 overweight baggage (> 25 kg)
50  SELECT *
51  FROM baggage
52  WHERE weight_in_kg > 25
53  ORDER BY weight_in_kg DESC
54  LIMIT 3;
55
56  -- 10. Youngest passenger(s), full name
57  SELECT first_name || ' ' || last_name AS full_name, date_of_birth
58  FROM passengers
59  WHERE date_of_birth = (SELECT MAX(date_of_birth) FROM passengers);
60
61  -- 11. Cheapest booking price on each booking platform
62  SELECT booking_platform, MIN(ticket_price) AS cheapest_price
63  FROM booking
64  GROUP BY booking_platform;
65
66  -- 12. Airlines whose airline_code contains a digit
67  SELECT *
68  FROM airline
69  WHERE airline_code ~ '[0-9]';
```

Below the query editor, there is a "Data Output" section with a table showing the results of the highlighted query (Query 10). The table has two columns: "full_name" (text) and "date_of_birth" (date). The first row shows "full_name" as "text" and "date_of_birth" as "date".

At the bottom of the IDE, a status bar shows "Total rows: 0" and "Query complete 00:00:01.527". A green notification bar at the bottom right states: "Successfully run. Total query runtime: 1 secs 527 msec. 0 rows affected."

11

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postgres/postgres@PostgreSQL 18

Query

Query History

AI Assistant

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72

73

74

75

LIMIT 3;

-- 10. Youngest passenger(s), full name

SELECT first_name || ' ' || last_name AS full_name, date_of_birth

FROM passengers

WHERE date_of_birth = (SELECT MAX(date_of_birth) FROM passengers);

-- 11. Cheapest booking price on each booking platform

SELECT booking_platform, MIN(ticket_price) AS cheapest_price

FROM booking

GROUP BY booking_platform;

-- 12. Airlines whose airline_code contains a digit

SELECT *

FROM airline

WHERE airline_code ~ '[0-9]';

-- 13. Top 5 most recently created airlines

SELECT *

FROM airline

ORDER BY created_at DESC, airline_id DESC

LIMIT 5;

Scratch Pad x

Data Output

Messages

Notifications

booking_platform

cheapest_price

character varying (50)

numeric

SQL

✓ Successfully run. Total query runtime: 1 secs 879 msec. 0 rows affected. ✕

Total rows: 0

Query complete 00:00:01.879

LF

Ln 61, Col 1

12

Welcome X postgres/postgres@PostgreSQL 18* X

postgres/postgres@PostgreSQL 18

Query

Query History

AI Assistant

Execute script
F5

54

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73

74

75

LIMIT 3;

-- 10. Youngest passenger(s), full name

SELECT first_name || ' ' || last_name AS full_name, date_of_birth

FROM passengers

WHERE date_of_birth = (SELECT MAX(date_of_birth) FROM passengers);

-- 11. Cheapest booking price on each booking platform

SELECT booking_platform, MIN(ticket_price) AS cheapest_price

FROM booking

GROUP BY booking_platform;

-- 12. Airlines whose airline_code contains a digit

SELECT *

FROM airline

WHERE airline_code ~ '[0-9]';

-- 13. Top 5 most recently created airlines

SELECT *

FROM airline

ORDER BY created_at DESC, airline_id DESC

LIMIT 5;

Scratch Pad X

Data Output

Messages

Notifications

Showing rows: 1 to 200

Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	107	AIR107	Airline 107	COUNTRY 7	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
2	108	AIR108	Airline 108	COUNTRY 8	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
3	109	AIR109	Airline 109	COUNTRY 9	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
4	110	AIR110	Airline 110	COUNTRY 10	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
5	111	AIR111	Airline 111	COUNTRY 11	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
6	112	AIR112	Airline 112	COUNTRY 12	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
7	113	AIR113	Airline 113	COUNTRY 13	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
8	114	AIR114	Airline 114	COUNTRY 14	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
9	115	AIR115	Airline 115	COUNTRY 15	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
10	116	AIR116	Airline 116	COUNTRY 16	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
11	117	AIR117	Airline 117	COUNTRY 17	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
12	118	AIR118	Airline 118	COUNTRY 18	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
13	119	AIR119	Airline 119	COUNTRY 19	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
14	120	AIR120	Airline 120	COUNTRY 20	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
15	121	AIR121	Airline 121	COUNTRY 1	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876

Total rows: 200

Query complete 00:00:01.223

Successfully run. Total query runtime: 1 secs 223 msec. 200 rows affected.

LF Ln 66, Col 1

13

Welcome × postgres/postgres@PostgreSQL 18* ×

postgres/postgres@PostgreSQL 18

Query

Query History

AI Assistant

Scratch Pad ×

```
61 -- 11. Cheapest booking price on each booking platform
62 SELECT booking_platform, MIN(ticket_price) AS cheapest_price
63 FROM booking
64 GROUP BY booking_platform;
65
66 -- 12. Airlines whose airline_code contains a digit
67 SELECT *
68 FROM airline
69 WHERE airline_code ~ '[0-9]';
70
71 -- 13. Top 5 most recently created airlines
72 SELECT *
73 FROM airline
74 ORDER BY created_at DESC, airline_id DESC
75 LIMIT 5;
76
77 -- 14. booking_id between 200 and 300 inclusive, check_result <> 'Checked'
78 SELECT *
79 FROM baggage_check
80 WHERE booking_id BETWEEN 200 AND 300
81 AND check_result <> 'Checked';
82
83 -- 15. update at in the same month as created at, but earlier than created at
```

Data Output

Messages

Notifications

Showing rows: 1 to 5

Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	204	FLY	FlyFly	POLAND	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
2	203	FLH	FlyHigh	BRAZIL	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
3	202	AEZ	AirEasy	FRANCE	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
4	201	KAZ	KazAir	TURKEY	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876
5	200	AIR200	Airline 200	COUNTRY 20	2026-09-29 13:12:12.628876	2026-09-29 13:12:12.628876

✓ Successfully run. Total query runtime: 1 secs 308 msec. 5 rows affected. ✕

Total rows: 5 | Query complete 00:00:01.308 | LF | Ln 71, Col 1

14

Welcome x postgres/postgres@PostgreSQL 18 x

postgres/postgres@PostgreSQL 18

Query Query History AI Assistant Execute script F5

```
66 -- 12. Airlines whose airline_code digit
67 SELECT *
68 FROM airline
69 WHERE airline_code ~ '[0-9]';
70
71 -- 13. Top 5 most recently created airlines
72 SELECT *
73 FROM airline
74 ORDER BY created_at DESC, airline_id DESC
75 LIMIT 5;
76
77 -- 14. booking_id between 200 and 300 inclusive, check_result <> 'Checked'
78 SELECT *
79 FROM baggage_check
80 WHERE booking_id BETWEEN 200 AND 300
81 AND check_result <> 'Checked';
82
83 -- 15. update_at in the same month as created_at, but earlier than created_at
84 SELECT *
85 FROM baggage_check
86 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
87 AND updated_at < created_at;
```

Data Output Messages Notifications

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer

Successfully run. Total query runtime: 1 secs 157 msec. 0 rows affected.

Total rows: 0 Query complete 00:00:01.157 LF Ln 77, Col 1

15

Welcome × postgres/postgres@PostgreSQL 18* ×

postgres/postgres@PostgreSQL 18

Query Query History AI Assistant

Scratch Pad ×

```
66 -- 12. Airlines whose airline_code contains a digit
67 SELECT *
68 FROM airline
69 WHERE airline_code ~ '[0-9]';
70
71 -- 13. Top 5 most recently created airlines
72 SELECT *
73 FROM airline
74 ORDER BY created_at DESC, airline_id DESC
75 LIMIT 5;
76
77 -- 14. booking_id between 200 and 300 inclusive, check_result <> 'Checked'
78 SELECT *
79 FROM baggage_check
80 WHERE booking_id BETWEEN 200 AND 300
81       AND check_result <> 'Checked';
82
83 -- 15. update_at in the same month as created_at, but earlier than created_at
84 SELECT *
85 FROM baggage_check
86 WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
87       AND updated_at < created_at;
```

Data Output Messages Notifications

baggage_check_id [PK] integer

check_result character varying (50)

created_at timestamp without time zone

updated_at timestamp without time zone

booking_id integer

passenger_id integer

Total rows: 0 Query complete 00:00:01.002

Successfully run. Total query runtime: 1 secs 2 msec. 0 rows affected.

LF Ln 83, Col 1